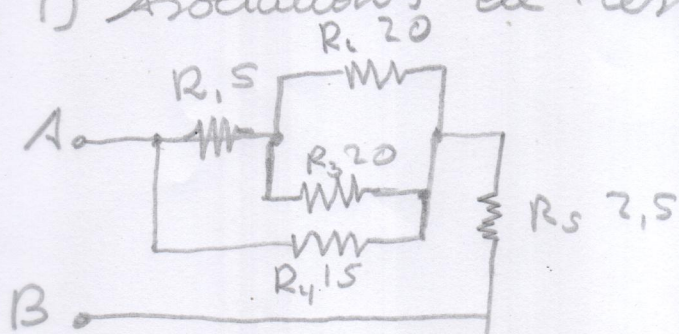


Aluno: Saul Astudillo ; CI = 33384958

1) Associações de resistências



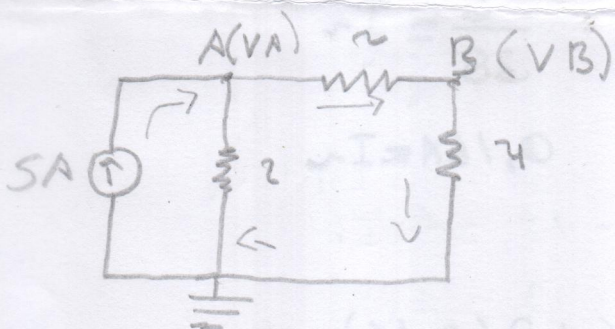
$$\frac{1}{R_{23}} = \frac{1}{20} + \frac{1}{20} = \frac{40}{20^2} \Rightarrow R_{23} = \frac{20^2}{40} = \frac{400}{40} = 10\Omega$$

$$R_{123} = 5 + 10 = 15\Omega$$

$$\frac{1}{R_{1234}} = \frac{1}{15} + \frac{1}{15} = \frac{30}{15^2} \Rightarrow R_{1234} = \frac{15^2}{30} = 7,5\Omega$$

$$R_{eq} = 7,5 + 2,5 = 10\Omega$$

2) Análise nodal



$$SA = \frac{VA}{2} + \frac{VA - VB}{2}$$

$$10A = VA + VA - VB$$

$$10A = 2VA - VB$$

$$10A = 2VB$$

$$5A = VB$$

$$2 \frac{(VA - VB)}{2} = \frac{VB}{4}$$

$$2VA - 2VB = VB$$

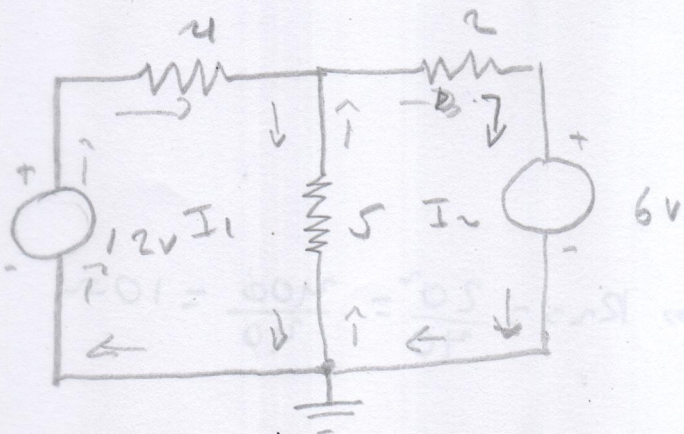
$$2VA - 3VB = 0$$

$$2VA = 3VB$$

$$VA = \frac{3VB}{2}$$

$$VA = 7,5A$$

3- Analisis de mallas



$$12V - 4I_1 - 5(I_1 - I_2) = 0$$

$$12V - 9I_1 + 5I_2 = 0$$

$$12V = 9I_1 - 5I_2$$

$$-6V - 5(I_2 - I_1) - 2(I_2) = 0$$

$$-6V - 5I_2 + 5I_1 - 2I_2 = 0$$

$$-6V = 7I_2 - 5I_1$$

$$-6 = -5I_1 + 7I_2$$

$$I_{sc} = 1,42 - 0,16$$

$$I_{sc} = 1,26 A$$

$$\begin{cases} 12V = 9I_1 - 5I_2 \times 5 \\ -6V = -5I_1 + 7I_2 \times 9 \end{cases}$$

$$60V = 45I_1 - 25I_2$$

$$-54V = -45I_1 + 63I_2$$

$$116 = 38I_2$$

$$\frac{6}{38} = I_2$$

$$0,16A \approx I_2$$

$$\frac{6 + 7(0,16)}{5} \approx I_1$$

$$1,42 \approx I_1$$